Poisson regression

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Warmup

Work on the warmup activity (handout)

Warmup

	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	0.276493	0.087294	3.167	0.00154	**
phd	0.008071	0.026358	0.306	0.75944	
ment	0.026537	0.002001	13.264	< 2e-16	***
kid5	-0.099696	0.035248	-2.828	0.00468	**

How would we interpret the coefficient on the number of articles published by the student's mentor?

Warmup

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What is the predicted number of articles for a student at a PhD program with a prestige score of 4, whose mentor published 30 articles in the last 3 years, and who has no children?

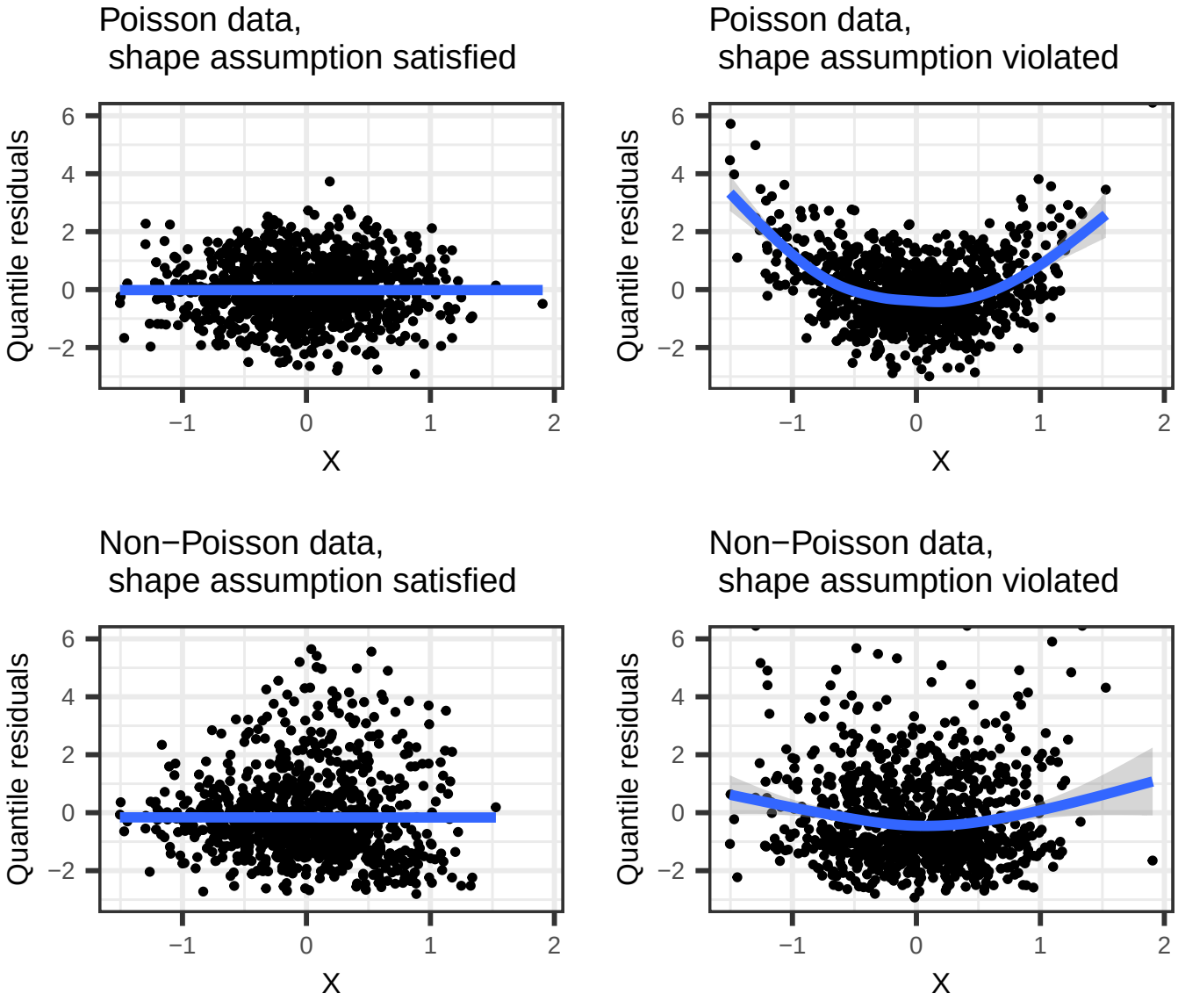
Model assumptions and diagnostics

- Empirical log mean plots
- Quantile residual plots
- QQ plots of the quantile residuals
- Rootograms

Activity

Work on the class activity (course website).

Class activity: Poisson regression assumptions



Using quantile residual plots

All assumptions are satisfied:

- Residuals are randomly scattered around 0, with no clear pattern
- Residuals have constant variance
- Residuals are mostly between -2 and 2

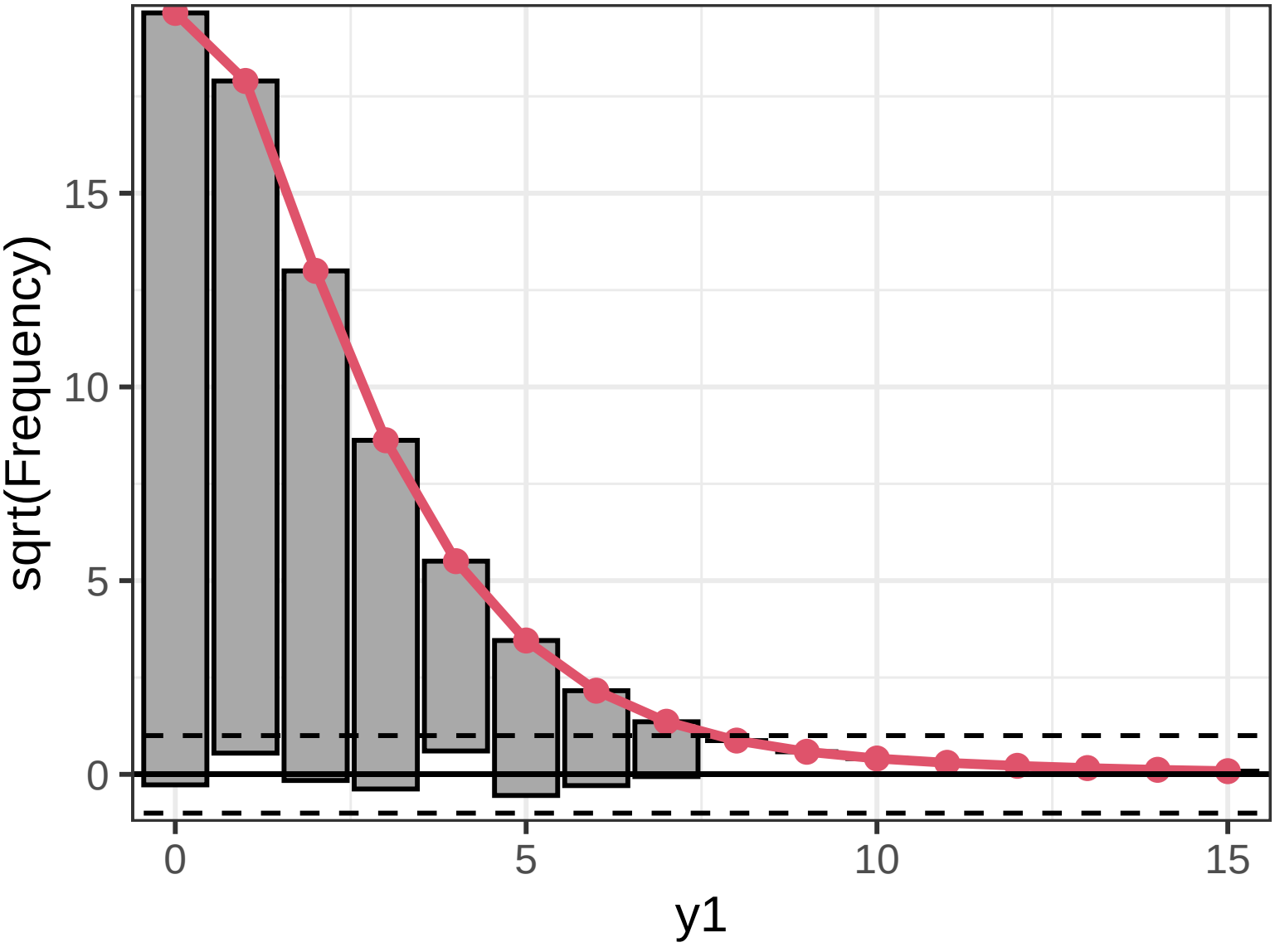
If you see a pattern in the residuals:

- Shape assumption may not be satisfied

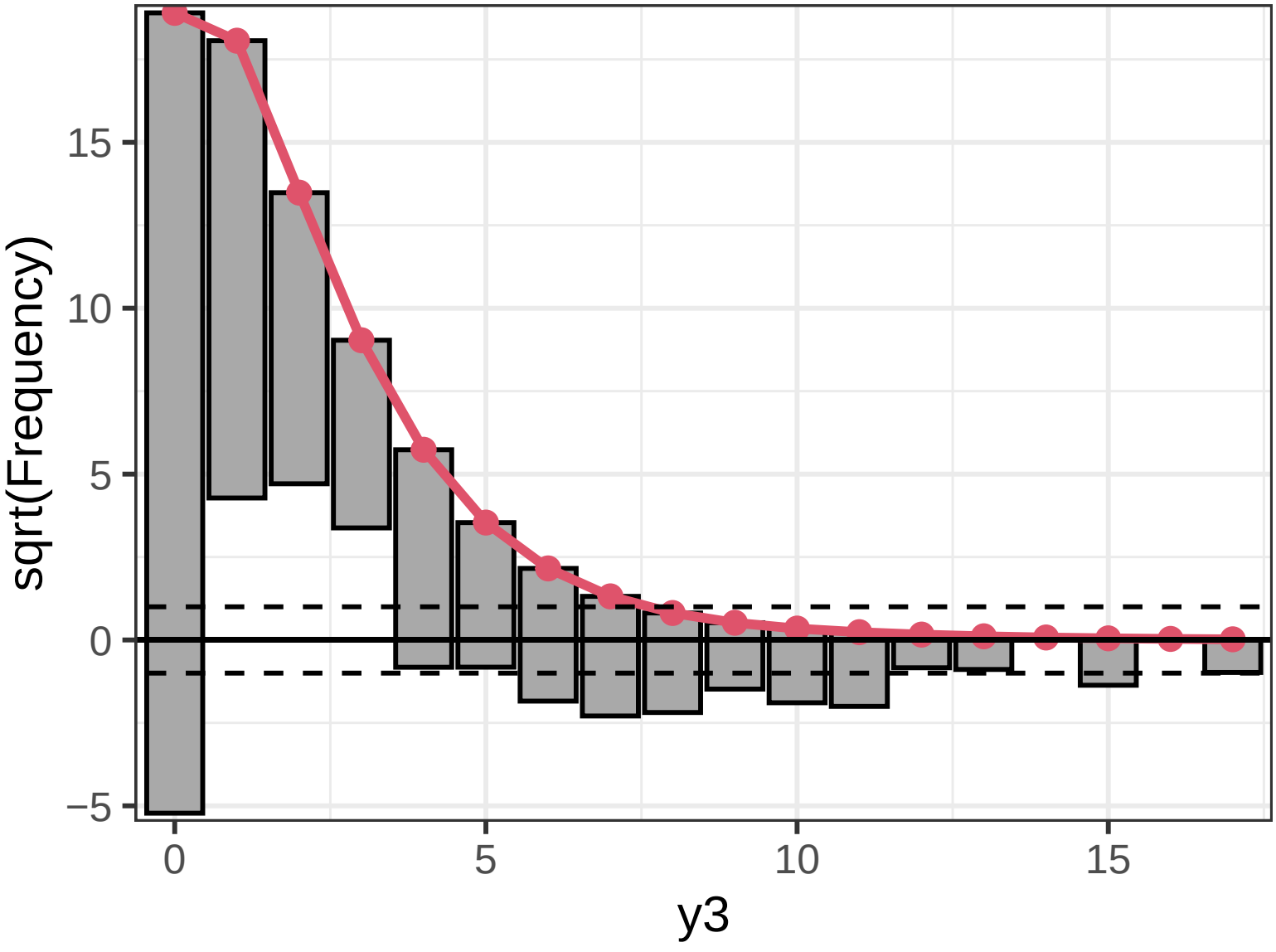
If residuals have non-constant variance, and/or variance is too large:

- Distribution assumption may not be satisfied

Rootograms

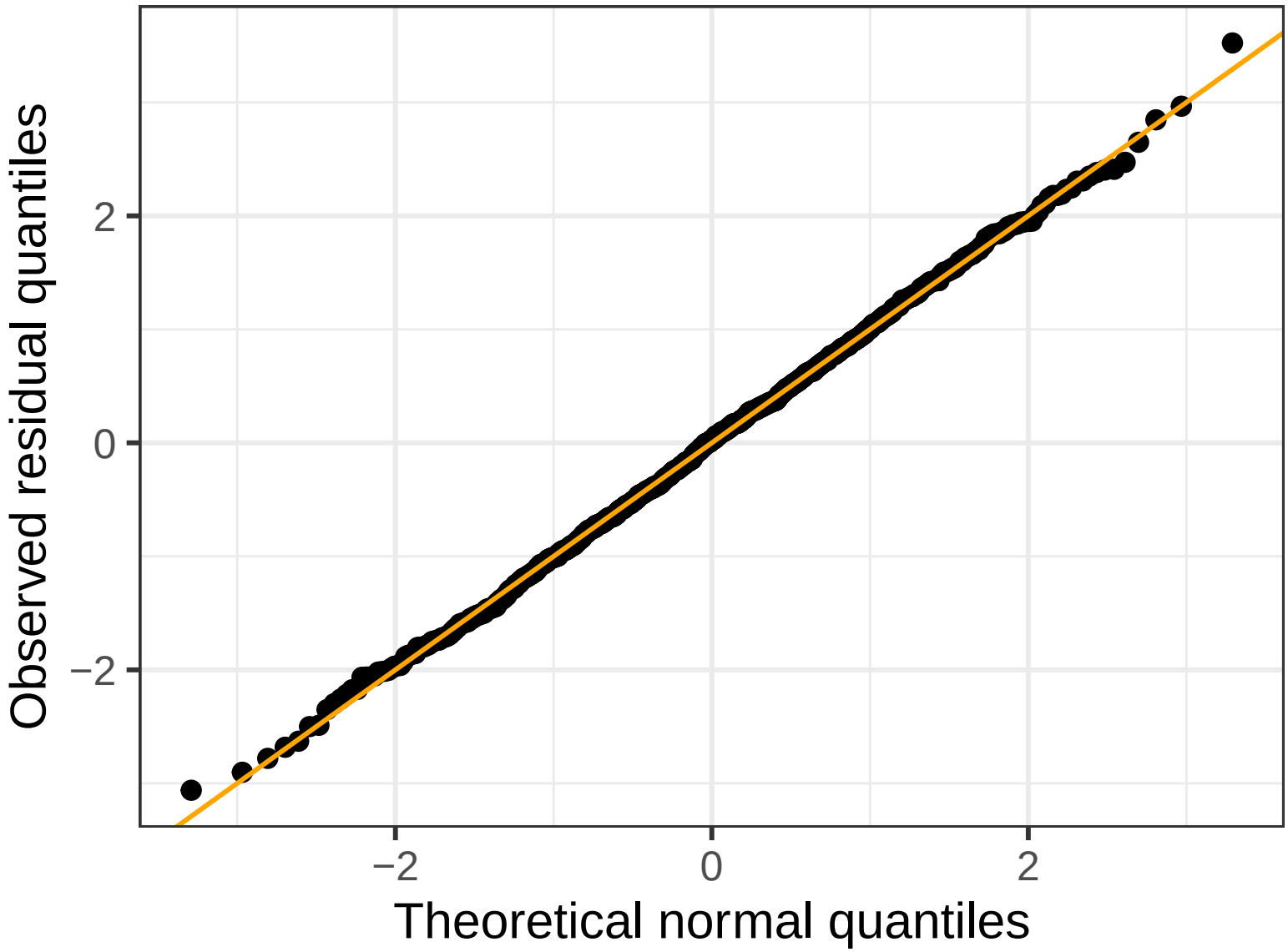


Examining rootograms



QQ plots

When the model assumptions are satisfied:



QQ plots

When the model assumptions are not satisfied:

